

Causal voltage relaxation after GITT pulses

The source contains measured Li-graphite half-cell GITT and POCV exports. This task uses the GITT campaign only. A completed current pulse is followed by a long rest; the practical question is whether two early rest-voltage observations can forecast the subsequent 30 minutes.

The strongest development-selected rule is finite-pulse diffusion, with no fitted global coefficient. Its reserved MAE is 0.686 mV, a 12.2% improvement over the constrained flexible control and 80.1% over persistence. The advantage is specific to MAE: its RMSE and worst-block error are worse than the flexible control. This is a useful scoped forecasting result, not a newly discovered universal electrochemical law.

Let $g(t)=\sqrt{(t+T)}-\sqrt{t}$. Eliminating the unknown amplitude and equilibrium offset from $U(t)=U_{\infty}+A\ g(t)$ gives $U(t)=U_b+(U_b-U_a)[g(t)-g(b)]/[g(b)-g(a)]$. Here T is the measured pulse duration and a, b are the actual causal calibration times. The cancellation explains both its simplicity and why it cannot identify a unique diffusivity.

Experimental scope and evaluation

Issue predictions at 120 s of each current-free block; only actual samples at or before 10/60/120 s and completed preceding current pulse are predictors. Targets are nearest native samples to 300/600/1200/1800 s, tolerance 15 s.

Calibration: Three causal voltage samples per block; no fit to later outcomes. Calibration access is identical for every model.

Validation unit: Complete rest block linked to preceding current pulse; all blocks belong to one half-cell, so block summaries are not independent-cell confidence.

Report chronological block errors; no independent-cell confidence interval.

Target: voltage during current-free relaxation (V). Errors use V. The selected reference is **finite_pulse_diffusion**. Selection was frozen before confirmation; diagnostic winners are not substituted afterward.

Predictor	Unit
t	s
t60	s
t120	s
u60	V
u120	V
u10	V
pulse_s	s
current	A

Matched comparison

Model	Development MAE	Reserved MAE	Reserved worst-group MAE
fixed_rc	0.00139811	0.00218012	0.0509115
flexible	0.00060865	0.000782191	0.0115079
persistence	0.00150566	0.00344433	0.100651
finite_pulse_diffusion	0.000391423	0.000686422	0.0144938
learned_rc	0.000670335	0.000993741	0.0151337
fractional_tail	0.000459116	0.000604582	0.00785862
diffusion_rc_mixture	0.000391423	0.000686422	0.0144938
boundary_storage	0.000392807	0.000568418	0.00734045

The primary error is mean group MAE. RMSE and signed bias are complementary; rows within a group do not establish independent replication. No accuracy percentage or industrial tolerance is invented.

Selected equation and coefficients

$$\widehat{U}(t) = U_b + (U_b - U_a) \frac{g(t) - g(b)}{g(b) - g(a)}, \qquad g(t) = \sqrt{t + T} - \sqrt{t}$$

T is completed pulse duration; a and b are actual calibration timestamps at or before 60 and 120 s. Voltages U_a and U_b are measured calibration values. No global coefficient is fitted.

This reference has no globally fitted coefficients.

All comparator expressions, numerical guards and full-precision values remain in EQUATIONS.md, rules.json and run.py.

Applicability and limitations

- One Li-graphite half-cell and one GITT campaign; no population transfer or diffusion-coefficient identification.
- POCV file is preserved but not used because it has a different protocol.

Only initial pre-GITT voltage rows were inspected at schema stage. Eligible reserved relaxation targets were not inspected. All packaged outcomes become public/exposed.

Reproduction

Run python run.py in this final package to verify hashes and reproduce every saved prediction. Use python run.py --inputs new.csv --output predictions.csv --model reference only with the declared inputs and units. Numerical prediction evaluation does not certify novelty or mechanism. Source anchors and reserved observations are separate from predictor inputs.

Source: <https://zenodo.org/records/17295469>. See source/units audit and prior-art records in research history.